Choosing the Right Test: Goodness-of-Fit Assessment in Censored Survival Data

A Simulation Study

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July, 2026

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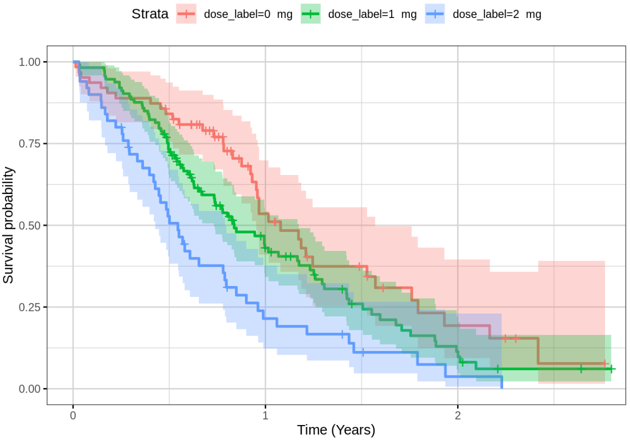

Research Question

Which goodness-of-fit test provides the most reliable performance for right-censored survival data?

1. Compare the performance of the Anderson-Darling, Cramér-von Mises and Kolmogorov-Smirnov tests under right censoring.

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2. Investigate the effect of sample size, skewness and censoring mechanism on the statistical power of each test.

Objectives of the Thesis

1. Compare the performance of the Anderson-Darling, Cramér-von Mises and Kolmogorov-Smirnov tests under right censoring.
2. Investigate the effect of sample size, skewness and censoring mechanism on the statistical power of each test.
3. Develop practical guidelines to support the selection of the most appropriate goodness-of-fit test in real applications.

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3. Develop practical guidelines to support the selection of the most appropriate goodness-of-fit test in real applications.

Main contribution

The study proposes the $Diff_{t_{\max}}$ diagnostic metric as an additional tool to improve test selection under right-censored data.

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Null hypothesis

$$H_0 : \; F(t) = F_0(t;\theta), \qquad \text{for some } \theta \in \Theta$$

Alternative hypothesis

$$H_1 : \; F(t) \neq F_0(t;\theta)$$

- Unknown parameters are estimated by MLE.
- Parameter estimation requires bootstrap calibration.

Main idea: The CvM test measures the overall discrepancy between the empirical and theoretical CDFs.

Test Statistic

$$\widehat{M}_n = n \int_{-\infty}^{+\infty} \left(\widehat{F}_n(t) - F_0(t; \hat{\theta}) \right)^2 \, dF_0(t)$$

Right-censored data The empirical CDF is replaced by the Kaplan–Meier estimator.

Key feature

Sensitive to discrepancies across the entire distribution.

Main idea The AD test compares the empirical and theoretical CDFs, emphasizing the distribution tails.

Test Statistic

$$\widehat{A}_n = n \int_{-\infty}^{+\infty} \frac{\left(\widehat{F}_n(t) - F_0(t;\hat{\theta})\right)^2}{F_0(t;\hat{\theta})\left(1 - F_0(t;\hat{\theta})\right)} \, dF_0(t;\hat{\theta})$$

Key feature

Most sensitive to tail discrepancies.

Main idea The KS test measures the maximum distance between the empirical and theoretical CDFs.

Test Statistic

$$D_n = \sup_t |F_n(t) - F_0(t)|$$

Unlike CvM and AD, the KS test is based on the maximum discrepancy rather than the overall area between the two CDFs.

Key feature

The KS test focuses exclusively on the **largest local discrepancy** between the empirical and theoretical distributions.

Key Differences

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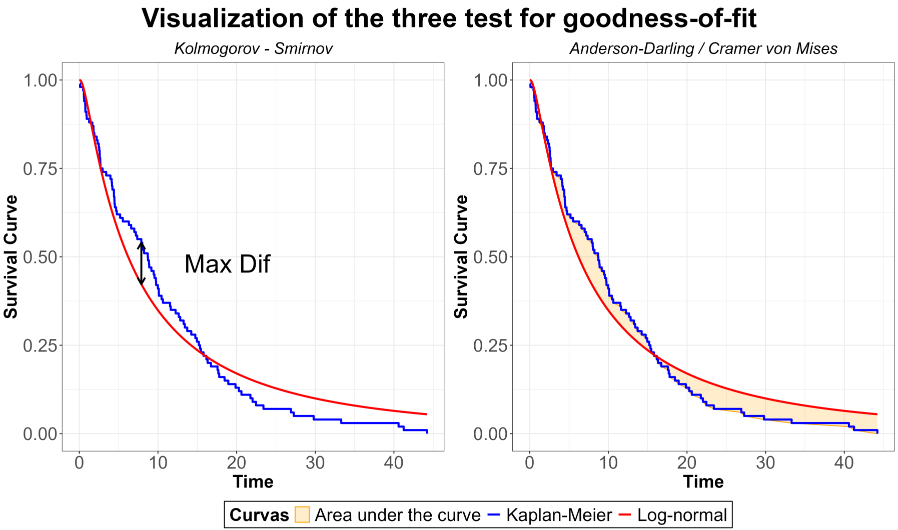

Problem

Unknown parameters are estimated from the same sample, leading to smaller test statistics and invalid classical critical values.

Solution

Conditional bootstrap:

1. Estimate parameters (MLE).
2. Generate bootstrap samples.
3. Re-estimate parameters.
4. Compute the empirical null distribution.

Key idea

Bootstrap preserves the nominal Type I error.

Simulation Scenarios

- **108 Scenarios** - 1000 Samples each

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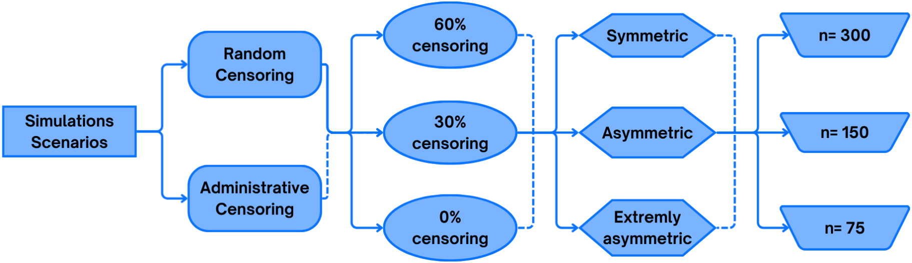

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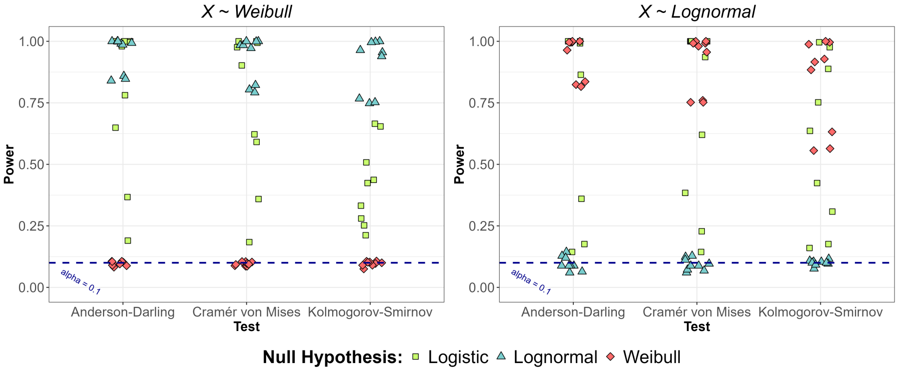
Complete Data:



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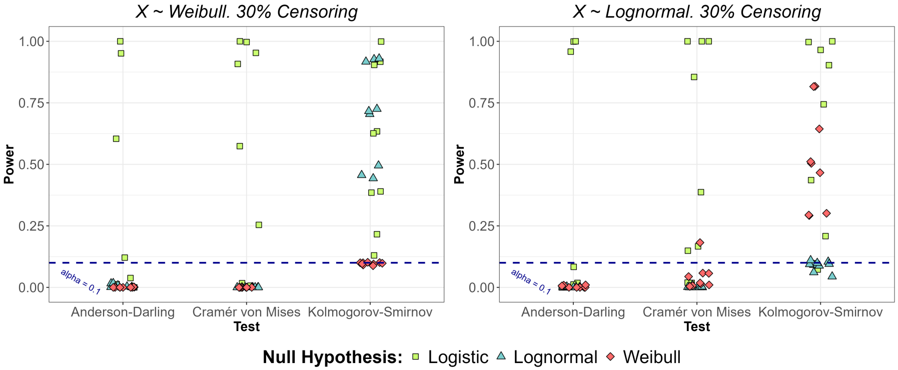
Administrative Censored Data:



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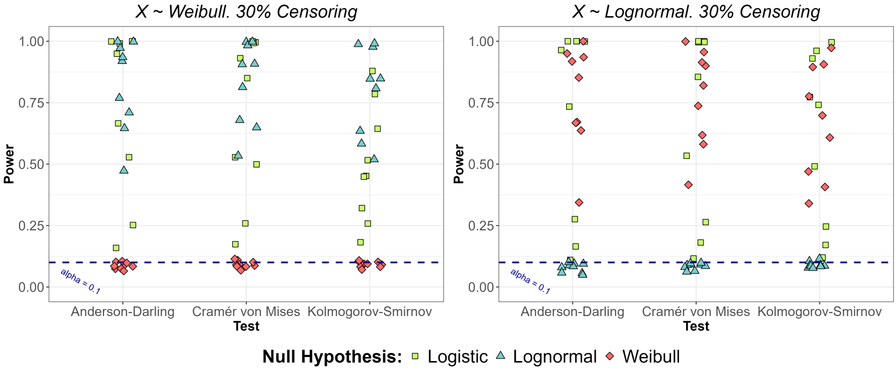
Random Censored Data:



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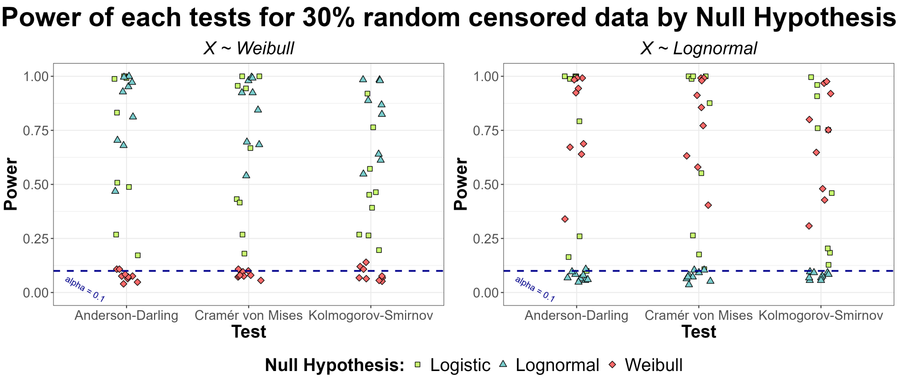

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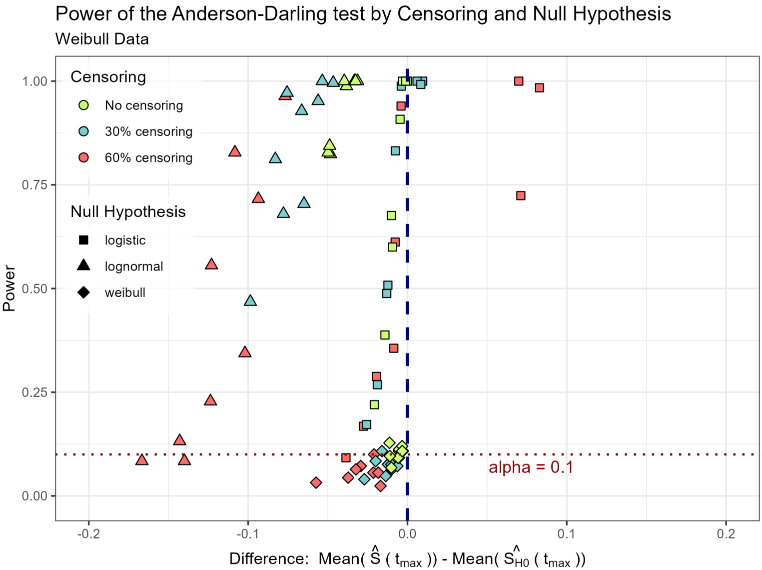

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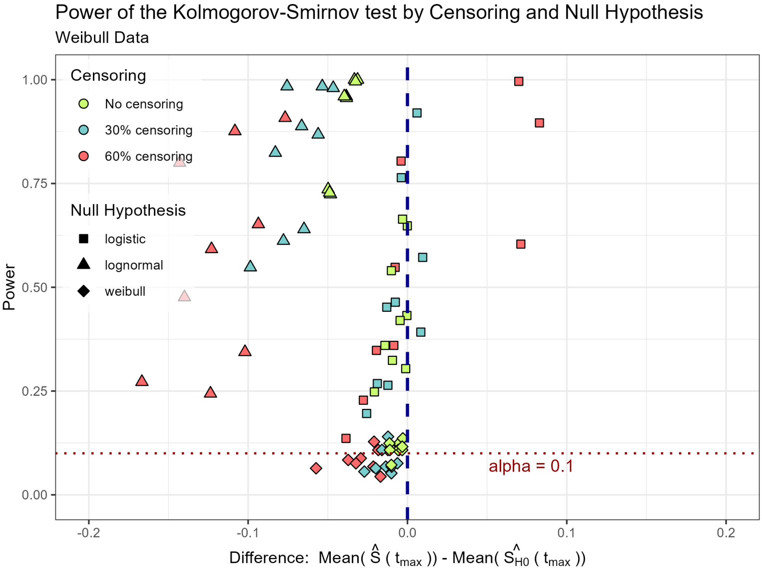

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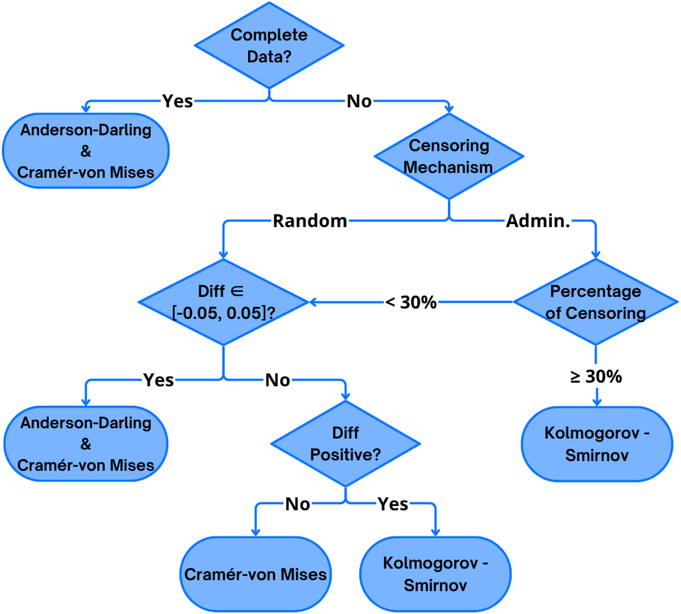

Thank you for your attention

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